Infinite Series Worksheet (9.3)

Name: _____ Date: _____

Type A: Geometric Series

Determine whether the series converges, and if so find its sum. Recall: $\sum_{k=0}^{\infty} ar^k = a + ar + ar^2 + \cdots + ar^k + \cdots$ converges to $\dfrac{a}{1-r}$ if $|r| < 1$ and diverges if $|r| \geq 1$. Some series are geometric in *concealed form* — rewrite them first.

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1. $\displaystyle\sum_{k=0}^{\infty} \frac{7}{3^k} = 7 + \frac{7}{3} + \frac{7}{3^2} + \cdots + \frac{7}{3^k} + \cdots$

☐ converges, sum = _____ ☐ diverges

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2. $\displaystyle\sum_{k=0}^{\infty} \frac{(-1)^k\, 6}{5^k} = 6 - \frac{6}{5} + \frac{6}{5^2} - \cdots + \frac{(-1)^k\, 6}{5^k} + \cdots$

☐ converges, sum = _____ ☐ diverges

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3. $\displaystyle\sum_{k=1}^{\infty} 2^{2k}\, 3^{1-k} = 4 + \frac{16}{3} + \frac{64}{9} + \cdots + 2^{2k}\, 3^{1-k} + \cdots$

☐ converges, sum = _____ ☐ diverges

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4. $\displaystyle\sum_{k=1}^{\infty} \frac{5^{k+1}}{8^k} = \frac{25}{8} + \frac{125}{64} + \frac{625}{512} + \cdots + \frac{5^{k+1}}{8^k} + \cdots$

☐ converges, sum = _____ ☐ diverges

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5. $\displaystyle\sum_{k=2}^{\infty} \frac{3^k}{2^{3k-1}} = \frac{9}{32} + \frac{27}{256} + \frac{81}{2048} + \cdots + \frac{3^k}{2^{3k-1}} + \cdots$

☐ converges, sum = _____ ☐ diverges

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6. $3 + \dfrac{3}{4} + \dfrac{3}{4^2} + \cdots + \dfrac{3}{4^{k-1}} + \cdots$

☐ converges, sum = _____ ☐ diverges

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7. $\dfrac{2}{9} - \dfrac{4}{27} + \dfrac{8}{81} - \cdots + \dfrac{(-1)^{k-1}\, 2^k}{3^{k+1}} + \cdots$

☐ converges, sum = _____ ☐ diverges

Type B: Telescoping Sums

Find a closed form for the nth partial sum s_n (use partial fractions or log rules), then determine whether the series converges by computing $\lim_{n\to+\infty} s_n$. If it converges, find its sum.

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8. $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^2+3k+2} = \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \cdots + \frac{1}{k^2+3k+2} + \cdots$

☐ converges, sum = _____ ☐ diverges

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9. $\displaystyle\sum_{k=1}^{\infty} \frac{1}{(2k-1)(2k+1)} = \frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} + \frac{1}{5\cdot 7} + \cdots + \frac{1}{(2k-1)(2k+1)} + \cdots$

☐ converges, sum = _____ ☐ diverges

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10. $\displaystyle\sum_{k=1}^{\infty} \ln\left(\frac{k}{k+1}\right) = \ln\frac{1}{2} + \ln\frac{2}{3} + \ln\frac{3}{4} + \cdots + \ln\left(\frac{k}{k+1}\right) + \cdots$

☐ converges, sum = _____ ☐ diverges

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11. $\displaystyle\sum_{k=1}^{\infty} \frac{2}{k(k+2)} = \frac{2}{1\cdot 3} + \frac{2}{2\cdot 4} + \frac{2}{3\cdot 5} + \cdots + \frac{2}{k(k+2)} + \cdots$

☐ converges, sum = _____ ☐ diverges

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12. $\displaystyle\sum_{k=1}^{\infty} \frac{4}{k^2+4k+3} = \frac{4}{2\cdot 4} + \frac{4}{3\cdot 5} + \frac{4}{4\cdot 6} + \cdots + \frac{4}{k^2+4k+3} + \cdots$

☐ converges, sum = _____ ☐ diverges

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13. $\displaystyle\frac{1}{4\cdot 5} + \frac{1}{5\cdot 6} + \frac{1}{6\cdot 7} + \cdots + \frac{1}{(k+3)(k+4)} + \cdots$

☐ converges, sum = _____ ☐ diverges

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14. $\displaystyle\frac{1}{1\cdot 4} + \frac{1}{4\cdot 7} + \frac{1}{7\cdot 10} + \cdots + \frac{1}{(3k-2)(3k+1)} + \cdots$

☐ converges, sum = _____ ☐ diverges